



# AI-based air quality PM<sub>2.5</sub> forecasting models for developing countries: A case study of Ho Chi Minh City, Vietnam

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## ABSTRACT

Outdoor air pollution damages the climate and causes many diseases, including cardiovascular diseases, respiratory infections, and lung damage. In particular, Particulate Matter (PM<sub>2.5</sub>) is considered a hazardous air pollutant to human health. Accurate hourly forecasting of PM<sub>2.5</sub> concentrations is thus of significant importance for public health, helping the citizens to plan the measures to alleviate the harmful effects of air pollution on health.

This study analyses and discusses the temporal characteristics of PM<sub>2.5</sub> at different locations in Ho Chi Minh City (HCMC), Vietnam - an economic center and a megacity in a developing country with a population of 8.99 million people. We developed several AI-based one-shot multi-step PM<sub>2.5</sub> forecasting models, with both an hourly forecast granularity (1 h to 24 h) and a 24-h rolling mean. These Machine Learning algorithms include Stochastic Gradient Descent Regressor, hybrid 1D CNN-LSTM, eXtreme Gradient Boosting Regressor, and Prophet. We collected the data from six monitoring stations installed by the HealthyAir project partners at different locations in HCMC, including traffic, residential and industrial areas in the city. In addition, we developed a suitable model training protocol using data from a short period to address the non-stationarity of PM<sub>2.5</sub> time series. Our proposed PM<sub>2.5</sub> forecasting models achieve state-of-the-art accuracy and will be deployed in our HealthyAir mobile app to warn HCMC citizens of air pollution issues in the city.

## 1. Introduction

Clean air is the foremost requirement to sustain human health and the environmental ecosystem. Air pollution has become a very serious problem in many countries which is silently impacting our health including the increase in numbers of heart, asthma, and lung cancer diseases (Xing et al., 2016; Kurt et al., 2016; Mannucci and Franchini, 2017; Bourdrel et al., 2017). Air pollution also creates environmental problems, such as global warming, acid rain, and the ozone layer's depletion. Day by day, air pollution increases due to uncontrolled traffic and poor industrial setups in urban areas (Danh, 2020). According to the World Health Organization (WHO), 9 out of 10 people live in a place where air pollution exceeds the WHO threshold (WHO, 2021). A study by Lelieveld et al. (2015) shows that

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outdoor air pollution, mainly due to PM<sub>2.5</sub>, leads to 3.3 million premature deaths per year globally, and this number will likely double by 2050 (Lelieveld et al., 2015).

In Vietnam, a study by WHO (WHO, 2018) reports >60 thousand deaths each year due to diseases linked to air pollution: heart, respiratory, cardiovascular, and pulmonary diseases. HCMC is the biggest city in Vietnam, with the highest population density of 4,476 inhabitants/km<sup>2</sup>; 80% of its population of 8.99 million lives in urban areas. Still, its PM<sub>2.5</sub> level is increasing due to Vietnam's growing economy and urbanisation; without additional environmental measures, the level of PM<sub>2.5</sub> in HCMC and other cities in Vietnam may increase by 30% by 2030 (Amann et al., 2019).

Besides policy changes, an immediate remedy for air pollution's harmful effects is to develop PM<sub>2.5</sub> forecasting models, whose results will allow the public to plan how to avoid dangerous air pollution periods in the near future. However, forecasting PM<sub>2.5</sub> has always been challenging due to its space-time variation and its dependency on many environmental factors. Various statistical models have been used to predict PM<sub>2.5</sub> concentrations (Mahajan et al., 2018; Zhang et al., 2018); still, the performance of statistical models is limited compared to AI-based PM<sub>2.5</sub> models when dealing with nonlinearity and time-varying data (Kim and Lee, 2021). Table 1 lists AI-based methods used in PM<sub>2.5</sub> forecasting and shows that PM<sub>2.5</sub> predictions results vary from different geolocations (such as Beijing, Shanghai, Tehran, HCMC) due to their spatiotemporal nature and weather conditions. Regarding the forecasting targets, hourly and multi-step (1h to 24h) PM<sub>2.5</sub> prediction represent a more complicated task than the daily averaged and single step (next day or next hour) PM<sub>2.5</sub> prediction.

A recent study in Vietnam developed daily averaged PM<sub>2.5</sub> forecasting models for HCM City using single-source data from the U.S. Consulate (Minh et al., 2021). Yet, a daily PM<sub>2.5</sub> forecasting model does not provide the required time granularity in their predictions for the citizens to plan their day ahead, specifically when high levels of air pollution occur at different periods during the day. Furthermore, a forecasting model for a single location is insufficient to represent the entire city due to the variation of PM<sub>2.5</sub> concentrations across different sites due to different levels of traffic, industrial and residential activity.

In this paper, we developed AI-based hourly PM<sub>2.5</sub> forecasting models to predict the hourly PM<sub>2.5</sub> concentrations in the upcoming 24 h across six locations in HCMC. We also developed models for 24-h rolling mean PM<sub>2.5</sub> forecasting, as recommended by the Vietnam National Technical Regulation on Ambient Air Quality QCVN 03:2015 (MONRE, 2013) and global air quality guidelines (WHO, 2021).

The main objectives of this study are: 1) Analyze the temporal characteristics of PM<sub>2.5</sub> across the HCM city. 2) Develop multiple machine learning models for hourly PM<sub>2.5</sub> forecasting and 24-h rolling mean PM<sub>2.5</sub> forecasting for different regions of HCMC, including traffic, residential and industrial areas, and 3) Assess the performance of various PM<sub>2.5</sub> forecasting models and find the most suitable model for deployment in our HealthyAir app - a mobile application developed by our team to inform the citizens of HCMC about the air pollution concentration in the city.

**Table 1**  
Summary of AI-based methods used in PM<sub>2.5</sub> forecasting model development.

| # | References                       | Study area                      | Study period | Sampling frequency | Prediction (forecasting horizon) | Methods & Results (RMSE)                                                                                                                                     |
|---|----------------------------------|---------------------------------|--------------|--------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Minh et al. (2021)               | HCMC, Vietnam                   | 2016 to 2019 | Daily mean         | next day and up to 7 days        | Linear Regression: (7.80)<br>Neural Network:(7.97<br>LASSO: (7.82)<br>Extra Trees Regression:(7.68)<br>Decision Forest Regression: (7.82)<br>XGBoost: (8.30) |
| 2 | Pak et al. (2020)                | Beijing, China                  | 2015 to 2017 | Daily mean         | next day's PM <sub>2.5</sub>     | MLP (37.792),<br>LSTM (11.337)<br>CNN-LSTM (5.287)                                                                                                           |
| 3 | Zamani Joharestani et al. (2019) | Tehran, Iran                    | 2015 to 2018 | Daily mean         | next day's PM <sub>2.5</sub>     | Random Forest (14.47)<br>XGBoost (13.66)<br>MLP (15.11)                                                                                                      |
| 4 | Zhang et al. (2018)              | Beijing, China                  | 2015         | 4 h (mean)         | next day's PM <sub>2.5</sub>     | Linear Regression (30.271)<br>Results for 24 h: Multiple Additive                                                                                            |
| 5 | Karimian et al. (2019)           | Tehran, Iran                    | 2013 to 2016 | Hourly             | 12 h, 24 h, 48 h                 | Regression Trees: 13.19<br>Deep Feed Forward Network: 19.62<br>LSTM: 9.42                                                                                    |
| 6 | Ma et al. (2020)                 | Shanghai, China                 | 2015 to 2018 | Hourly             | 24 h, 48 h                       | XGBoost (26.1)<br>Lasso regression (28.1)                                                                                                                    |
| 7 | Xiao et al. (2020)               | Beijing, Tianjin, Hebei - China | 2015 to 2017 | Daily mean         | next day's PM <sub>2.5</sub>     | WLSTME: 40.67                                                                                                                                                |
| 8 | Li et al. (2020)                 | US Embassy, Beijing             | 2010 to 2014 | Daily mean         | 24 h                             | Multivariate CNN-LSTM: 17.93<br>Multivariate LSTM: 18.08<br>Univariate CNN-LSTM: 19.77<br>Univariate LSTM: 23.26                                             |
| 9 | Zhong et al., 2021               | National stations across China  | 2016 to 2019 | Hourly             | Not mentioned                    | LightGBM: 19.80                                                                                                                                              |

### 1.1. The healthyair project

The two-year project started in June 2020, funded by the Irish Research Council and the Department of Foreign Affairs in Ireland through the COALESCE programme (Better World Awards 2020) for reducing humanitarian needs, climate action, and strengthening governance. The project aims to monitor and predict air pollution and its impact in climate change and human health, as well as to develop a tool which can help inform citizens as well as develop propose policy for HCM City to reduce air pollution and improve population health. The project is a collaboration between CeADAR, Ireland's Centre for Applied Artificial Intelligence, based in University College Dublin (UCD), Ireland, and Vietnam National University (VNU) in Ho Chi Minh City throughout the Institute for Environment and Resources and the School of Economics and Law.

## 2. Data and methods

### 2.1. Air quality monitoring network

The air quality monitoring network was established by the HealthyAir project team in February 2021. The air quality monitoring network consists of six operating stations (Table 2) across the HCM City (Fig. 1). Therefore, the period of data used in this research comprehend February 2021 to December 2021.

Each air quality monitoring station has dedicated sensors to monitor and record every minute concentrations of particulate pollutants, gaseous, and meteorological parameters in real-time. All the stations send the collected data to a central storage server which can be used for analysis and modeling. Each operating station is well equipped with:

- Dust sensors to measure the concentration of Particulate Matter (PM<sub>2.5</sub>) and Total Suspended Particles (TSP)
- Air pollution gas sensors including Sulfur Dioxide (SO<sub>2</sub>), Ozone (O<sub>3</sub>), Nitrogen Dioxide (NO<sub>2</sub>), Carbon Monoxide (CO)
- Sensors to measure the concentration of temperature and humidity (Rh)

### 2.2. Data pre-processing

First, the entire dataset was retrieved from a remote centralised server where all station data was stored in real-time. The air pollution data was recorded at 1-min sampling time and then the data was aggregated (mean) to 1 sample every hour.

A systematic inspection of our dataset showed that at times the sensors read unrealistically high or low values, e.g. values of PM<sub>2.5</sub> higher than 20000  $\mu\text{g}/\text{m}^3$  or smaller than  $-10 \mu\text{g}/\text{m}^3$ . We thus defined the valid range of values for hourly PM<sub>2.5</sub> to be between 0  $\mu\text{g}/\text{m}^3$  and 500  $\mu\text{g}/\text{m}^3$ . A value of hourly PM<sub>2.5</sub> concentration outside the defined valid range was replaced with a “nan” value, i.e. it was considered a missing value. We also identified outliers to be removed as those abnormal values of hourly PM<sub>2.5</sub> concentrations which are likely caused by unexpected events: i.e. hourly PM<sub>2.5</sub> values which are both higher than 200  $\mu\text{g}/\text{m}^3$  and that are three times higher than the hourly PM<sub>2.5</sub> values of the previous hours. The missing data at a particular hour was replaced with the concentration value of the previous hour. The missing values found in several consecutive hours were replaced with the concentration values of the same hours on the previous day. Next, our pre-processing procedure transformed each PM<sub>2.5</sub> time series into a supervised learning dataset of pairs of input/target. Our algorithm recorded the list of pairs whose target contains an imputed PM<sub>2.5</sub> concentration value to ensure only pairs without imputed hourly PM<sub>2.5</sub> value were used to evaluate a trained forecasting model.

The dataset was divided into training, validation, and testing sets in the proportion of 80:10:10. The training dataset was used to train the model, and the validation dataset was used to find the best hyperparameters settings of the models. The performance matrix of the model was calculated using the testing dataset. Finally, the data normalization was carried out to prepare the data on a small scale, we used min-max (Eq. 1) normalization to rescale the data in a range between 0 and 1.

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (1)$$

where,  $x'$  is the scaled feature (value between 0 and 1) and  $x$  is the original feature

**Table 2**

Air pollution monitoring stations and their location.

| Station # | Station category                                 | Locations                                                                            |
|-----------|--------------------------------------------------|--------------------------------------------------------------------------------------|
| 1         | Urban background: Industry+ Traffic +Residential | Vietnam national university in Ho Chi Minh city, Linh Trung ward, Thu Duc city, HCMC |
| 2         | Traffic                                          | 20 Nguyễn Trọng Trí street, An Lac ward, Binh Tan district, HCMC                     |
| 3         | Industry                                         | Tan Binh industrial zone/park, Tay Thanh ward, Tan Phu District, HCMC                |
| 4         | Residential                                      | 49 Thanh Da street, Ward 27, Binh Thanh District, HCMC                               |
| 5         | Traffic                                          | 268 Nguyen Dinh Chieu street, ward 6, District 3, HCMC                               |
| 6         | Traffic + Residential                            | MM18 Truong Son street, ward 14, District 10. HCMC                                   |

Note: All air quality stations monitor the same list of pollutants which includes PM<sub>2.5</sub>, TSP, NO<sub>2</sub>, SO<sub>2</sub>, O<sub>3</sub>, CO, and two meteorological parameters temperature and humidity.

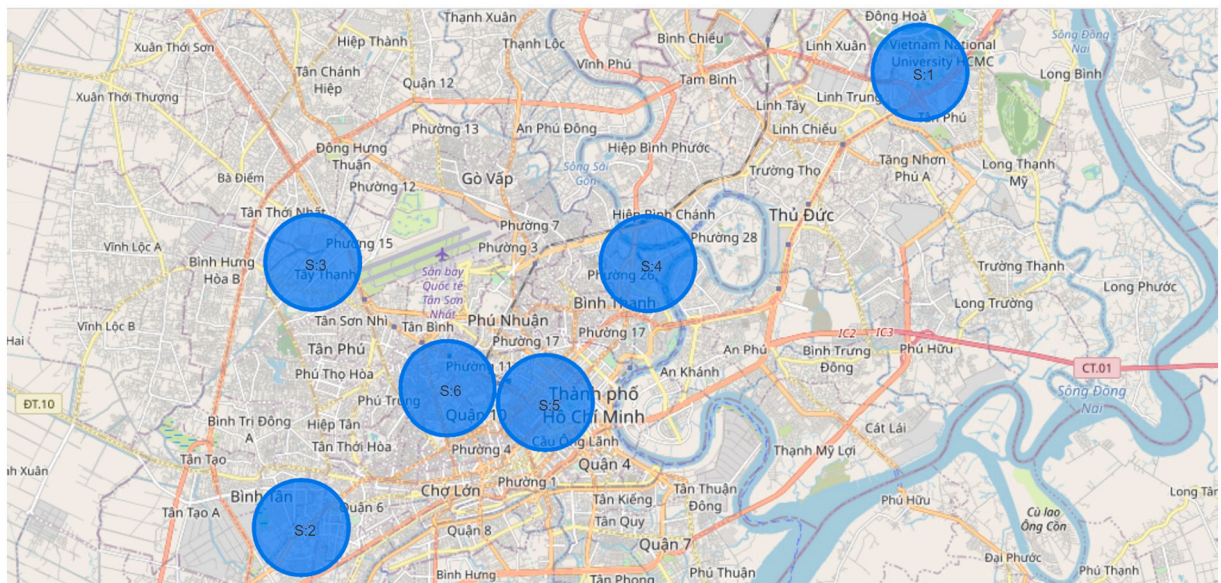


Fig. 1. Air pollution monitoring stations across HCMC.

### 2.3. PM<sub>2.5</sub> forecasting model development

In order to implement a one-shot multistep model for PM<sub>2.5</sub> prediction, the input and output data were prepared in a supervised learning fashion, where each pair of the input-output samples consists of the past 60 h of PM<sub>2.5</sub> observations as input and the next 24 h (a list of 1h to 24h) of data as output. As an example, if the current timestamp is time ( $t$ ) then the single pair of input series  $X$  should be  $[t - 59, t - 58, t - 57, \dots, t - 1, t]$  and the output series  $y$  would be  $[t + 1, t + 2, t + 3, \dots, t + 24]$  and the next pair of input series will be  $[t - 58, t - 57, t - 56, \dots, t, t + 1]$  and the output series will be  $[t + 2, t + 3, t + 4, \dots, t + 25]$ , and so on. In this study, empirically we found that the input length of 60 h is an optimal length of input in our case. To choose the optimal length, the experiments were conducted using a set of input lengths including [12h, 24h, 36h, 48h, 60h, 72h, 84h, 96h].

Several machine learning algorithms can be used for time series forecasting. This study considered three fast machine learning models for PM<sub>2.5</sub> forecasting - XGBoost (Chen and Guestrin, 2016), SGDR regressor (Bottou, 2010), and hybrid 1D CNN-LSTM. We also used Prophet (Taylor and Letham, 2018) to train a model for forecasting tasks comparison purposes.

The SGDR regressor and XGBoost methods don't directly support multistep continuous output to forecast 1 h to 24-h future observations. We have used a wrapper of multi-output regressor which takes as input of XGBoost or SGD regressor and trains the model using a multi-output regressor object to produce multi-step output.

#### 2.3.1. XGBoost Regressor

XGBoost is a supervised learning method which accepts training data composed of one or more features ( $x_i$ ) to predict a target variable ( $y_i$ ) using the mathematical structure given in the following Eq. 2.

$$\hat{y}_i = \sum_j \theta_j x_{ij} \text{ where } \theta \text{ is used to denote the model parameters} \quad (2)$$

The main task of training the model is to find the best hyperparameters ( $\theta$ ) that best fit the training data ( $x_i$ ) to the target variable ( $y_i$ ) using the optimized objective function. The objective function consists of a regularization term ( $\theta$ ) and training loss  $L(\theta)$  to measure how well the model fits on training examples, which is defined in Eq. 3.

**Table 3**  
Hyper parameters settings (XGBoost).

| # | Parameters       | Settings (values) |
|---|------------------|-------------------|
| 1 | n_estimators     | 100               |
| 2 | max_depth        | 5                 |
| 3 | learning_rate    | 0.1               |
| 4 | min_child_weight | 6                 |
| 5 | colsample_bytree | 0.5               |
| 6 | booster          | gbtree            |

$$\text{objective}(\theta) = L(\theta) + \Omega(\theta) \quad (3)$$

here, we have used the mean squared error loss  $L(\theta)$  function to calculate the error rate of the model. The formation of the objective function and further explanation are given in Appendix A. Based on our experiments the optimal hyperparameters selected for model development are shown in Table 3.

Initially, grid search was performed and then manually hyperparameters tuning was carried out to choose the best parameter settings for all proposed models.

### 2.3.2. SGD Regressor

The SGD regressor used gradient descent to iteratively find the minimum possible value of the cost function. In each iteration, SGDR approximates the true gradient of  $E(w, b)$  by considering a single training example at a time, instead of a sum of gradients for all the training examples.

Due to the randomness in descent, SGDR takes a higher number of iterations to reach the minima but still it is computationally less expensive. The loss function and regularization terms (penalties) have been used to fit the regression model (Eq. 4).

$$E(w, b) = \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i)) + \alpha R(w) \quad (4)$$

where  $L$  is a loss function,  $R$  is the regularization term (L2 norm) and  $\alpha > 0$  hyperparameters are used to control the regularization strength. We have used the Least-Squares as a loss function to compute the total squared error between actual data and the associated hyperplane.  $L$  and  $R$  are defined in Equation-5 and Equation-6 respectively.

$$L(y_i, f(x_i)) = \frac{1}{2} (y_i - f(x_i))^2 \quad (5)$$

$$R(w) := \frac{1}{2} \sum_{j=1}^m w_j^2 = \|w\|_2^2 \quad (6)$$

The hyperparameters used in SGDR for PM<sub>2.5</sub> forecasting model development are shown in Table 4.

### 2.3.3. Hybrid neural network (1D CNN-LSTM)

The LSTM (Long-Short Term Memory network) is an RNN (Recurrent Neural Network) based architecture of deep learning that is widely used in processing sequential data such as time series forecast modeling. The model learns the function that maps a sequence of time series observations to the desired forecasting outputs. The components of an LSTM architecture consist of a cell, a forget gate, an input, and an output gate. The interactive operations among the gates help LSTM to solve the long-term dependency problem that simple RNN cannot learn.

Here, we used a hybrid neural network instead of using a pure LSTM where a one-dimensional Convolutional Neural Network (CNN) is put on top of the LSTM network. The first part of the network (1D CNN) is used to automatically learn the salient features and the second part of the network (LSTM) is used to analyze the features executed by 1D CNN and estimate PM<sub>2.5</sub> concentrations for the next 24 h. The architecture and the hyperparameters used in the model are depicted in Fig. 2.

### 2.3.4. Prophet forecasting model

Prophet (Taylor and Letham, 2018) is a decomposable time series forecasting model built on an additive model where non-linear trends are fit with different seasonal periods. It works robustly with time-series data that have strong seasonal effects. For the two main components of the model, we estimated the growth component  $g(t)$  using piecewise linear growth and the seasonal component  $s(t)$  using a partial Fourier sum (Eq. 7). We did not consider the public holidays in our Prophet models.

$$\text{forecast } y(t) = g(t) + s(t) + \epsilon_t \quad (7)$$

where  $\epsilon_t$  is the error term (unusual changes not considered by the model).

This model is robust in handling missing values and outliers by fitting them with trend changes (Appendix B). Therefore, we did not handle missing values during model training in Prophet. The model hyperparameter tuning was carried out using time series cross-validation. To avoid model overfitting and underfitting problems in Prophet, we have set an optimal value of 0.1 to adjust the

**Table 4**  
Hyper Parameters settings (SGDRegressor).

| # | Parameters                  | Settings (values)         |
|---|-----------------------------|---------------------------|
| 1 | loss                        | squared_error             |
| 2 | eta0(initial learning rate) | 0.02                      |
| 3 | alpha                       | 0.003                     |
| 4 | penalty                     | l2 (standard regularizer) |
| 5 | max_iterint                 | 800                       |



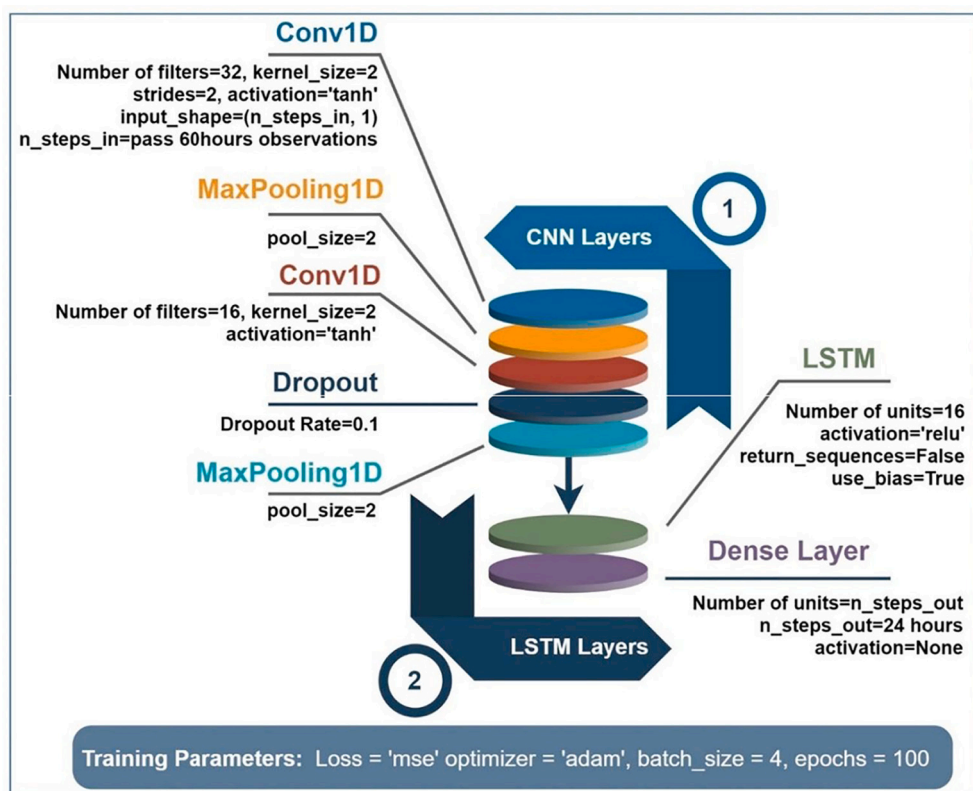


Fig. 2. 1D CNN-LSTM architecture with model hyperparameter settings.

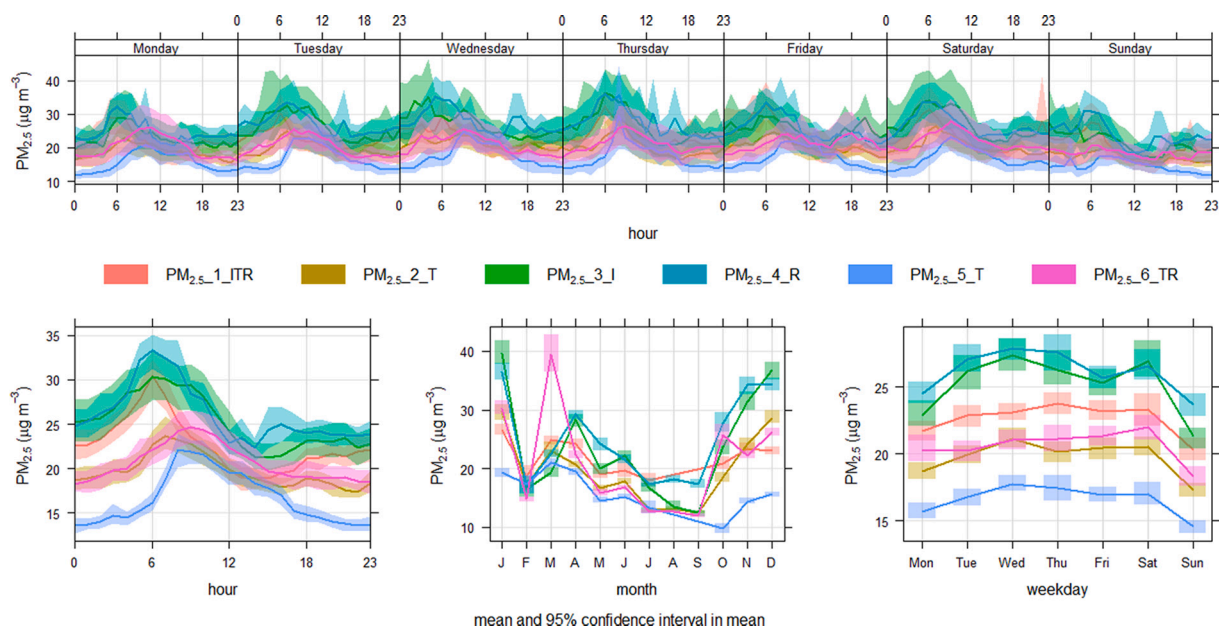
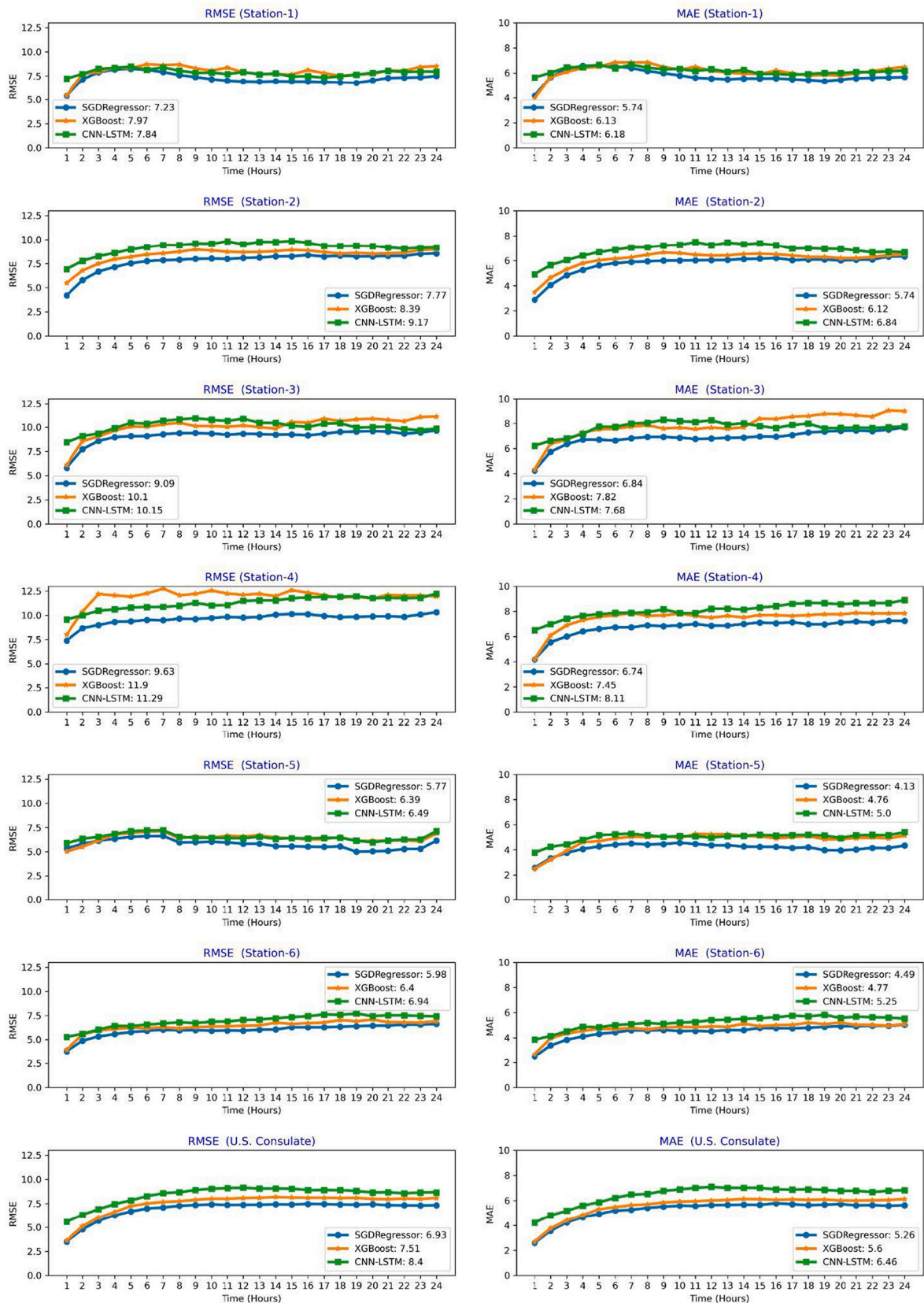


Fig. 3. Spatiotemporal characteristic of  $PM_{2.5}$  across the HCM City.



(caption on next page)

**Fig. 4.** Raw PM<sub>2.5</sub> forecasting results for all six stations + US Consulate.**Table 5**Raw PM<sub>2.5</sub> forecasting across all stations [averaged results].

| Evaluation matrix | XGBoost | 1D CNN-LSTM | SGD Regressor | Prophet |
|-------------------|---------|-------------|---------------|---------|
| RMSE              | 8.73    | 8.86        | 7.74          | 8.47    |
| MAE               | 6.28    | 6.65        | 5.75          | 5.89    |

strength of the sparse prior change points and a prior scale of 5.0 to adjust the changing strength of seasonality.

*Note: Data transformation in a supervised learning fashion and data scaling was not applied in the Prophet model as it has built-in functionality for scaling.*

### 2.3.5. Performance evaluation

The main task in prediction modeling is to approximate a mapping function ( $f$ ) from input variable ( $X$ ) to continuous output variable ( $y$ ). We seek to assess how close the prediction values are to the actual values. Therefore, the following performance metrics were used to calculate the error rates of our forecasting models:

We calculated the Root Mean Squared Error (RMSE) to estimate the average error rate of predicted values vs actual values, which is a square root of the average of square error across all samples of testing data.

$$\text{RMSE}(y, \hat{y}) = \frac{\sum_{i=1}^{n_{\text{testing\_samples}}} (y_i - \hat{y}_i)^2}{n_{\text{testing\_samples}}} \quad (8)$$

We also calculated the Mean Absolute Error (MAE) which takes the average absolute error between predictions and actual observations as a measurement of the magnitude of the error. The MAE is less sensitive to outliers than RMSE (Hyndman and Koehler, 2006). Therefore, the trends of RMSE and MAE might be different in some models.

$$\text{MAE}(y, \hat{y}) = \frac{1}{n_{\text{testing\_samples}}} \sum_{i=0}^{n_{\text{testing\_samples}}-1} |y_i - \hat{y}_i| \quad (9)$$

where,  $y_i$  indicates actual observations and  $\hat{y}_i$  represents predicted values

## 3. Results and discussion

### 3.1. Spatiotemporal characteristics and visualization – PM<sub>2.5</sub>

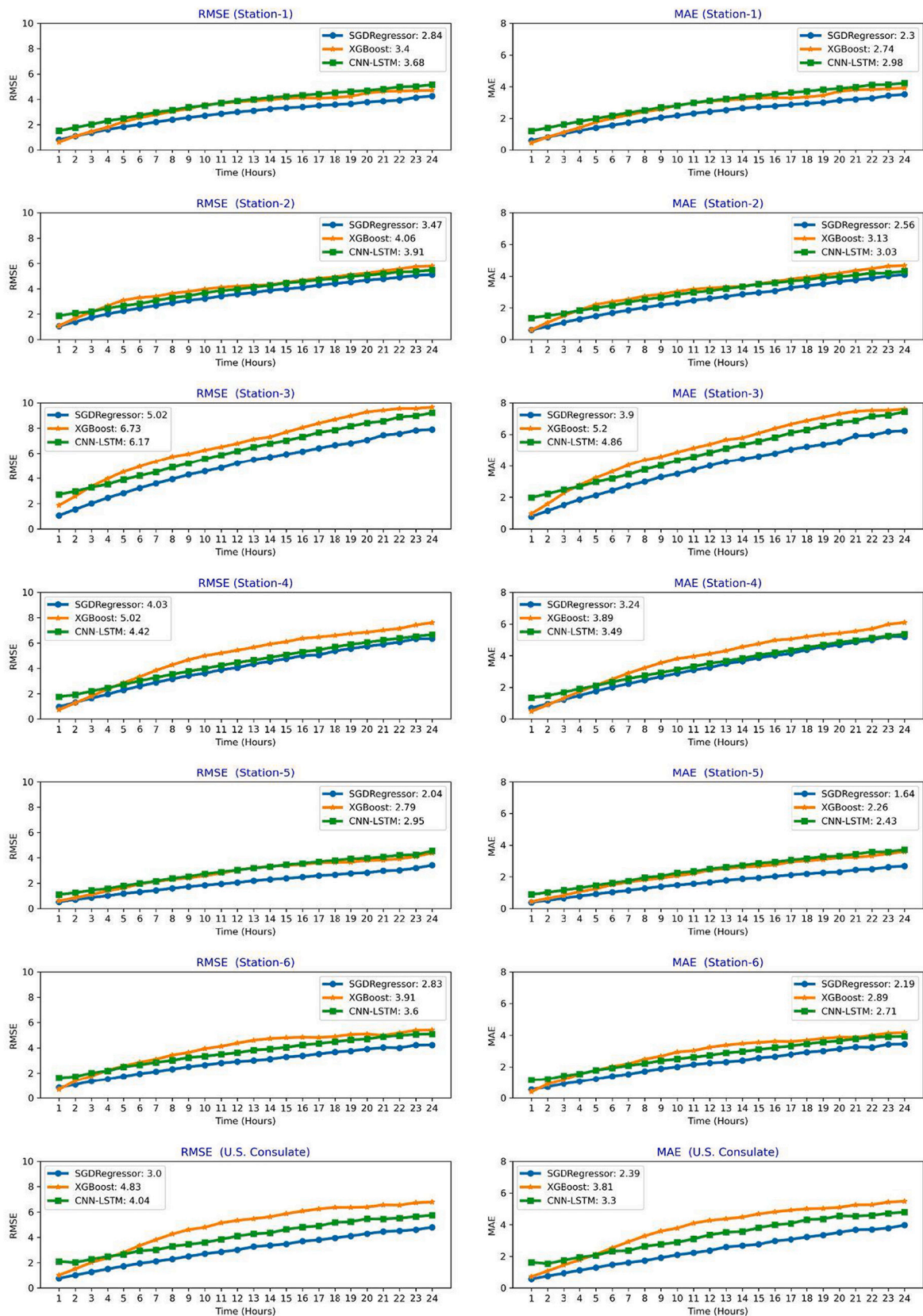
The WHO Ambient Air Quality Guideline (WHO, 2021) is set at 15  $\mu\text{g}/\text{m}^3$  for a 24 h average (with 99th percentile which is 3–4 exceedance days per year) and 5  $\mu\text{g}/\text{m}^3$  for an annual average.

For the annual average concentration of PM<sub>2.5</sub>, the highest concentration of 23.1  $\mu\text{g}/\text{m}^3$  is found at station 4, the reasons might be that this station is in a dense residential area and the residents in this area are poor, so generally they use wood, charcoal, and gas for cooking and they burn solid waste in open air. This annual average concentrations of PM<sub>2.5</sub> is about 4.6 times higher than the recommended by the WHO AQG. The second-highest level of the annual average PM<sub>2.5</sub> is found at station 3; this station is located in the Tan Binh industrial zone where many factories release a large amount of air pollution. The third highest polluted station is station 1, as this station is in a busy area with industrial activities from nearby Thu Duc city, Binh Duong province, and Dong Nai provinces. In addition, station 1 is the nearest station to the national highway, namely Xa Lo Ha Noi, and is also close to residential areas.

In general, in HCMC, the air pollution level is high from September to April and is low between May to August. This is due to the monsoon in HCMC, which contributes to reduce air pollution. However, 2021 has been an exceptional year due to the Covid-19 lockdown. A partial lockdown reduced about 50% of activities in HCMC from May to August, and a total lockdown happened in HCMC during the period of August to September. These lockdowns had an impact on the trends of PM<sub>2.5</sub> during the year 2021 (Fig. 3). The concentrations of PM<sub>2.5</sub> varied across different months; however, the highest concentration is found at almost all the stations during the winter season (November, December, January).

The highest pollution level was observed on Wednesday and Thursday. In HCMC public organizations do not work during the weekend which may help reduce air pollution. The level of PM<sub>2.5</sub> is found to be higher on Saturday as compared to Friday; one of the reasons might be due to the fact that many citizens leave the city on Saturday by motorcycles which leads to a higher level of pollution on Saturday. Fig. 3 also shows that the level of PM<sub>2.5</sub> is higher at 6–8 AM; the reasons might be that people go to work and cooking activities happen during that time. The level of PM<sub>2.5</sub> is also found to be higher at 5–6 PM when people return from work and cooking starts in households.





(caption on next page)

Fig. 5. 24 h rolling mean PM<sub>2.5</sub> forecasting results for all six stations + US Consulate.

Table 6

24 h rolling PM<sub>2.5</sub> forecasting across all stations [averaged results].

| Evaluation matrix | XGBoost | 1D CNN-LSTM | SGD Regressor | Prophet |
|-------------------|---------|-------------|---------------|---------|
| RMSE              | 4.31    | 4.12        | <b>3.38</b>   | 5.55    |
| MAE               | 3.36    | 3.25        | <b>2.64</b>   | 4.21    |

### 3.2. PM<sub>2.5</sub> forecasting results

In this study, various machine learning models were trained using hourly (1h, 2h, ..., 24h) raw PM<sub>2.5</sub> forecasting and 24-h rolling mean PM<sub>2.5</sub> forecasting.

In all the experiments, we split each dataset into train/validation/test data with the ratio (80%, 10%, 10%) along the time axis to avoid information leaks. Each model was trained on the train split, its hyperparameters were determined using the validation split, and its performance was calculated on the unseen test split. In addition, we used regularization to avoid overfitting. For these experiments we have collected and analysed data of PM<sub>2.5</sub> concentration for the year 2021, and considering that PM<sub>2.5</sub> time series are highly non-stationary and we do not have data from previous years, the data close to a chosen timestamp has been more relevant for forecasting PM<sub>2.5</sub> values. Therefore, for each station, instead of using the entire one-year dataset, we used uniform sampling to generate multiple datasets where the duration of each dataset is about three months. This protocol ensured that the training data was close to the test data in time for each generated dataset. Each dataset was then split into train/validation/test subsets to develop PM<sub>2.5</sub> forecasting models.

Fig. 4 depicts the average hourly performance of the hourly PM<sub>2.5</sub> forecasting models for each station. Our experiments show that the location of each station has a significant impact on the model performance, as the results vary from one station to another. Along the time dimension, the error rates (RMSE, MAE) are lower during the starting hours (h1, h2, h3), and then they increase gradually with time (except for station 5). Due to the critical power failure during the lockdown period, station 5 recorded missing data with large consecutive gaps.

For comparison purposes, we also carried out experiments on hourly PM<sub>2.5</sub> observations obtained from the US Consulate monitoring station (10.78335695 ° N, 106.6993907 ° E), which is closest to our HealthyAir station-5 (10.77636612 ° N, 106.6878094 ° E). The US consulate data covers all seasonal periods of the year. The results obtained from the U.S. Consulate data are in line with the results obtained from the data of our six monitoring stations (Fig. 4).

The average results of the hourly PM<sub>2.5</sub> forecasting models across all the stations for the three fast machine learning algorithms plus the Prophet models are presented in Table 5. The SGDRegressor models performed better than all other proposed models; its average error rate in terms of RMSE and MAE were 7.74 µg/m<sup>3</sup> and 5.75 µg/m<sup>3</sup>, respectively. It should be noted that each training set is quite small (80% of the total number of hourly examples of a three-month time series), which could explain the inferior performance of complex forecasting models like XGBoost or CNN-LSTM. In general Prophet works best on time series with strong seasonality, thus, Prophet models did not perform as well on our highly non-stationary air pollution time series. Still, the obtained results are competitive compared to existing PM<sub>2.5</sub> forecasting models (Table 1) for hourly PM<sub>2.5</sub> forecasting.

For the second task of 24 h rolling mean PM<sub>2.5</sub> forecasting, our models error rates for different forecasting hours are much lower than the error rates of raw hourly PM<sub>2.5</sub> forecasting (Fig. 5). This is because the rolling mean operation reduces the data variation over time and converts PM<sub>2.5</sub> time series data into a more linear form. The average error rates reported across all the stations for the three fast machine learning models plus the Prophet model (Table 6) show that the SGDRegressor performed better than other proposed models, with the lowest average RMSE of 3.38 µg/m<sup>3</sup> and MAE of 2.64 µg/m<sup>3</sup>.

## 4. Conclusion and future work

Air pollution is a severe problem that silently impacts our health. The situation is more urgent for large cities in developing countries. These cities are overpopulated and have to prioritise economic development over environmental protection. HCMC represents such a case; its population is 8.99 million people and it only had installed one air quality station before 2021. The HealthyAir project is an effort to systematically address HCMC's air pollution problems. The HealthyAir team installed six air quality stations to be able to record, analyze and predict air quality data in HCMC. This paper has analysed and discussed the spatiotemporal characteristics of air quality (PM<sub>2.5</sub>) across HCMC (Vietnam) and presented the series of machine learning models trained for raw hourly PM<sub>2.5</sub> forecasting and 24-h rolling mean PM<sub>2.5</sub> forecasting. To address the non-stationarity of PM<sub>2.5</sub> time series, we devised a new model training protocol: our system developed each forecasting model using data from a short period of about three months, keeping training data close to test data in time. We trained the forecasting models for each of the six stations in our air quality monitoring network,

covering different regions, including traffic, residential, and industrial areas. Our experiments show that the SGDRegressor consistently performed better than all other proposed models in both raw hourly PM<sub>2.5</sub> forecasting and 24-h rolling mean forecasting; in particular, it performed better than the popular Prophet model. Our proposed SGDRegressor models achieved state-of-the-art accuracy, so we plan to deploy them in the HealthyAir mobile app to provide accurate forecasts to the citizens of HCMC. We anticipate that our hourly raw PM<sub>2.5</sub> forecasting will be of great significance for the citizens in HCMC as it can provide early warning of dangerous air pollution hours during the day, while the 24 h rolling mean forecasting can be useful to understand the trend of PM<sub>2.5</sub> over the day. For future experiments, we plan to improve our PM<sub>2.5</sub> forecasting models by including meteorological and other air pollutant data. Also, once our air quality stations have recorded air pollution data for more than one year where lockdown periods will not occur, we will explore different protocols of training forecasting models on a much longer period of time to exploit long term seasonality patterns in the data.

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## Declaration of Competing Interest

None.

## Data availability

Data will be made available on request.

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## Appendix A. XGBoost Regressor - Formulation of Objective function used in PM<sub>2.5</sub> forecasting model development

The loss function used in model development is defined in Eq. A.1.

$$L(\theta) = \sum_i (y_i - \hat{y}_i)^2 \quad (\text{A.1})$$

Regularization is used to avoid model overfitting problems by controlling the complexity of the model. XGBoost uses a tree-based structure and leaf score for learning tasks. To learn the tree-based structure, let the Eq. A.2 be an objective function:

$$\text{objective}^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t)}) + \sum_{i=1}^t \Omega(f_i) \quad (\text{A.2})$$

where,  $\Omega$  penalize the complexity of model.

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (\text{A.3})$$

Learning tree-based structures is more complex than a traditional approach where a simple gradient is calculated. Learning all trees at once is more complex and needs more computational power. Instead, model training was carried out in an additive manner where new trees will be added at a time which optimize our objective. If the prediction value at time  $t$  is  $y_i^{(t)}$  then we have (Eq. A.4):

$$\hat{y}_i^{(t)} = \sum_{k=1}^t f_k(x_i) = \hat{y}_i^{(t-1)} + f_t(x_i) \quad (\text{A.4})$$

## Appendix B. Prophet - Handling missing values with trends and change points

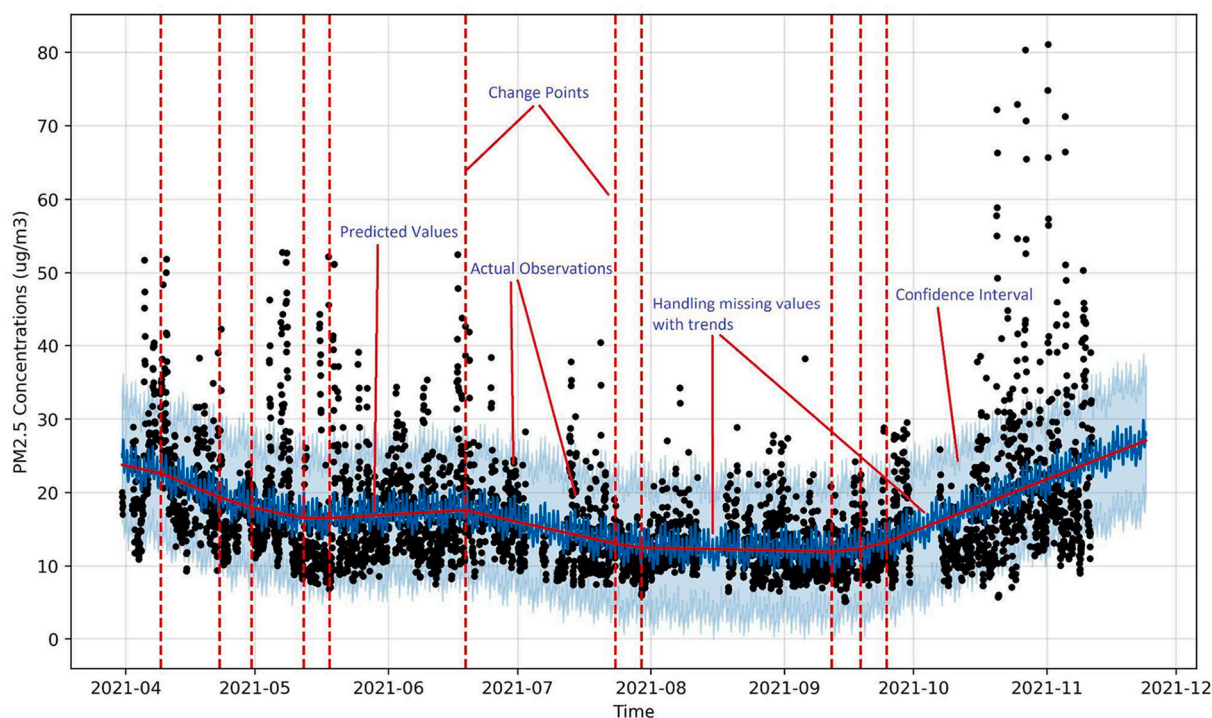


Fig. B.1. Prophet model for PM<sub>2.5</sub> forecasting with change points.

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